

Look Where You Work: View-Aligned Asset Identification with UWB and Smart Glasses

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Abstract

In dense industrial workspaces, operators using smart glasses often need asset-specific information for the machine they are currently observing. This is difficult because indoor positioning alone does not reliably determine the relevant asset when several nearby machines remain plausible candidates. As a result, the system may know the operator position, but not which workstation is actually being attended to. We present a preliminary prototype combining Ultra Wideband (UWB) positioning with on-device head-orientation sensing on commodity Android smart glasses. After a short heading calibration, the prototype fuses position and viewing direction to infer the most likely observed workstation and retrieves the corresponding device data via MQTT (Message Queuing Telemetry Transport). The goal is to show whether the concept works and where it breaks down, not to deliver a production-ready system. We evaluate the prototype in a controlled 3D-printer laboratory using a Leica laser-tracker reference system, static and dynamic motion experiments, and an exploratory practical identification test. Orientation-based disambiguation works under favorable conditions but fails for slow head rotations on all tested devices, which is the main open problem. In the practical test, the prototype correctly identified all four workstations when head rotations were fast enough. We release all data and source code artifacts as open source.

Keywords

Ultra-Wideband (UWB), Augmented Reality (AR), Smart Factory, Industrial IoT, Context-Aware Interfaces, Human-Machine Interaction, Location-Based Services

1. Introduction

Modern shop floors in smart factories are increasingly instrumented, yet many relevant tasks still take place under hands-busy conditions, for example while carrying parts, handling tools, or working in constrained spaces. In such situations, operators may need asset-specific information for the machine they are currently observing without interrupting their work to navigate menus or manually select a device. Smart glasses are a promising interface for this purpose because they provide hands-free information directly in the user's field of view [1]. However, indoor positioning alone is often insufficient in dense layouts, since user position does not indicate which nearby asset is currently being observed. Reliable context selection therefore requires an additional spatial cue such as head orientation or viewing direction [2].

A practical solution is to combine smart glasses with indoor positioning in a low-cost architecture [3] and to extend this setup with gyroscope-based head-orientation sensing. Ultra-wideband (UWB) is a well-established technology for indoor localization in industrial environments [6, 4, 5], while gyroscopes provide a practical signal for estimating head rotation on wearable devices, although drift and device-dependent stability remain relevant challenges [7]. By combining position and head orientation, asset selection can be aligned not only with the worker location but also with the direction the user is facing.

This paper makes the following contributions:

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- It identifies *asset disambiguation in dense indoor industrial layouts* as a limitation of position-only AR context selection and motivates head orientation as an additional spatial cue.
- It presents a *low-cost smart-glasses architecture* that combines UWB positioning, on-device orientation sensing, and MQTT-based integration for workstation-level context inference.
- It provides a *preliminary reference-based evaluation* showing that orientation-based selection works at sufficient rotation speeds but fails for slow head movements on all tested devices.
- It releases the complete Android implementation and all measurement data as open-source artifacts.

The remainder of the paper is structured as follows. Section 2 reviews related work. Section 3 outlines the system architecture. Section 4 describes the smart-glasses application, and Section 5 presents pose estimation and context inference. Section 6 covers the evaluation including laboratory setup and measurement results. Section 7 concludes the paper.

2. Related Work

Smart glasses and AR have been studied as hands-free interfaces for industrial assistance, including task guidance, maintenance support, and mobile information access [1, 8, 9]. UWB is well established for indoor positioning in industrial settings, with prior work on ranging methods, non-line-of-sight effects, and Industry 4.0 applications [6, 4, 5], as well as combinations with head-mounted AR displays [11]. Gyroscope-based head orientation has been used for target selection on wearables, though IMU estimates are affected by drift and device-dependent sensor fusion [7, 13]. Flatt et al. [10] combine AR and indoor localization for maintenance, but do not address asset disambiguation in dense layouts.

Our earlier work, *Step Into Context* [3], introduced a low-cost architecture for location-aware AR feedback based on UWB and MQTT, using proximity for context selection. This paper implements and evaluates orientation-based disambiguation for the case where proximity alone is ambiguous.

3. System Overview

Our system extends the low-cost architecture of [3] with orientation-aware context selection on smart glasses. Figure 1 summarizes the relevant data flow. Shop-floor devices publish telemetry via MQTT; a Raspberry Pi connected to a Qorvo DWM1001 UWB module publishes operator position estimates to the same broker.

On the operator side, a unified Android application subscribes to both UWB position data and device telemetry. After a brief user-triggered heading calibration, it estimates head orientation from on-device inertial sensors using relative heading updates to reduce magnetic sensitivity. A context-inference module fuses filtered position and head orientation to select the most likely observed workstation and renders the corresponding information in a view-aligned AR interface.

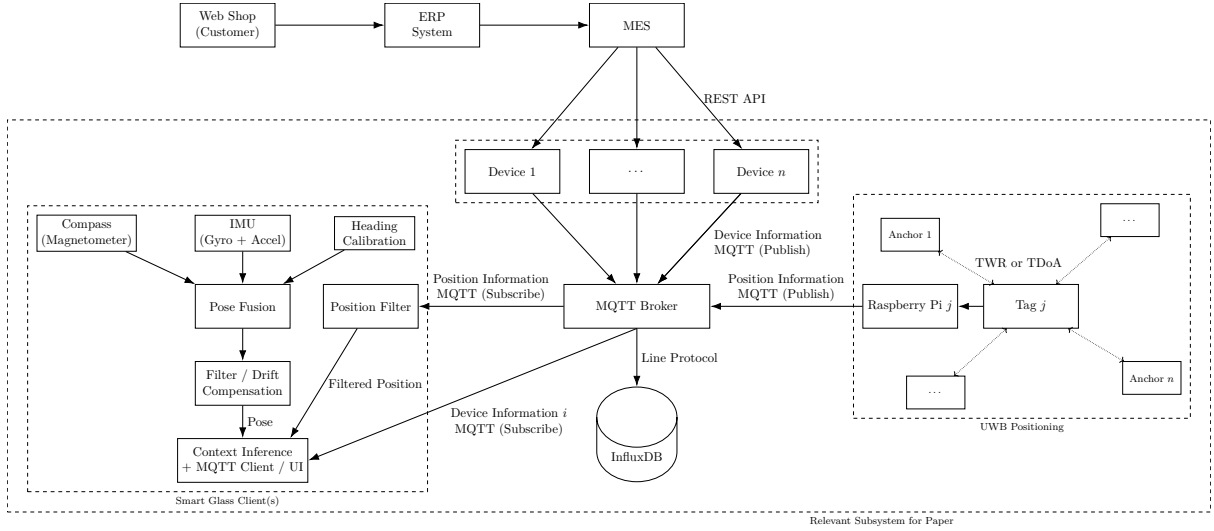


Figure 1: System overview. Orders are processed by ERP and MES and assigned to shop-floor devices, which publish status data via MQTT. A UWB setup provides operator position estimates. The smart-glasses app combines position and calibrated head orientation for workstation-level context inference and renders the selected device information in the AR interface.

4. Smart Glasses

In industrial environments, smart glasses act as hands-free user interfaces that present context-dependent instructions, live video, and machine data in the operator’s field of view.



Figure 2: Smart glasses used in the laboratory setup: Vuzix Blade 2 (top), Google Glass Enterprise Edition 2 (middle), and Vuzix M4000 (bottom).

A single Android application is deployed on Google Glass Enterprise Edition 2, Vuzix M4000, and Vuzix Blade 2 (Fig. 2); a Google Pixel 8 Pro serves as a conventional smartphone baseline. All three smart-glasses models run Android, provide a see-through display, and include a 3-axis gyroscope, accelerometer, and magnetometer. However, vendor documentation reports no gyroscope quality metrics such as drift rate, bias stability, or noise density, so sensor quality must be assessed experimentally. Although these are older models, direct contemporary replacements for Android-based smart glasses with integrated displays suitable for custom AR applications remain limited.

5. Pose Estimation and 2D Context Inference

The smart-glasses application estimates the wearer pose using the Android sensor framework. Orientation is obtained from the fused rotation estimate provided by the **Android SensorManager**, based on inertial measurements from the gyroscope and accelerometer. The yaw angle is used as the primary output, since workstation disambiguation is performed in a 2D floor-plane model.

UWB positioning data are received via MQTT and filtered on-device to suppress outliers and jitter. For this purpose, an α -trimmed filter is applied over a sliding window of recent position samples by discarding the largest and smallest α fraction and averaging the remaining values. This yields a stable position estimate at low latency.

The spatial coordinates of all candidate assets are stored in the application configuration and loaded at runtime. On application startup, the user performs a short heading calibration by facing a predefined reference direction. This defines the offset between the device-reported yaw and the global coordinate frame used for the device map.

Given the filtered user position $\mathbf{p} \in \mathbb{R}^2$ and the calibrated heading θ , context inference selects the most likely observed device in 2D. For each device with known position $\mathbf{d}_i \in \mathbb{R}^2$, the bearing angle is computed as

$$\phi_i = \text{atan2}((\mathbf{d}_i - \mathbf{p})_y, (\mathbf{d}_i - \mathbf{p})_x),$$

with wrapped angular difference

$$\Delta_i = \text{wrapToPi}(\phi_i - \theta).$$

The selected device is the candidate with minimal absolute angular deviation $|\Delta_i|$ within a fixed field-of-view threshold $\Delta_{\max}$. In the rare case of equal angular deviation, the nearer device is selected.

6. Evaluation

This section evaluates the proposed system from four perspectives: laboratory geometry, UWB position accuracy, rotational-motion response, and an exploratory practical identification test. A static 5-minute measurement confirmed that all devices maintain stable orientation estimates at rest (yaw standard deviation below 0.35° for all devices), so the following experiments focus on dynamic conditions.

6.1. Laboratory Setup

Figure 3 shows the laboratory environment used in the experiments. Four fixed reference points, labeled *A* to *D*, define the workspace geometry. The evaluation considers four 3D printers (footprint approximately 50×50 cm), denoted as p_A to p_D . The printers p_A , p_B , and p_C are arranged in one row, while p_D is placed separately. Each printer is represented by the center of its footprint in a 2D floor-plane model, with coordinates estimated using a laser tracker.

Figures 3a and 3b illustrate the two main geometric cases of the proposed context inference. In the unambiguous case, the viewing direction leads to a single dominant candidate, so the minimum-angle rule selects p_A directly. In the ambiguous case, several devices lie within the field of view and show similar angular deviations; the system resolves the ambiguity by selecting the nearest candidate, again yielding p_A .

6.2. Experimental Setup

A custom 3D-printed multi-device holder (Fig. 4) allowed three smart glasses, one smartphone, and the UWB tag to be mounted and moved together, ensuring all devices experienced identical

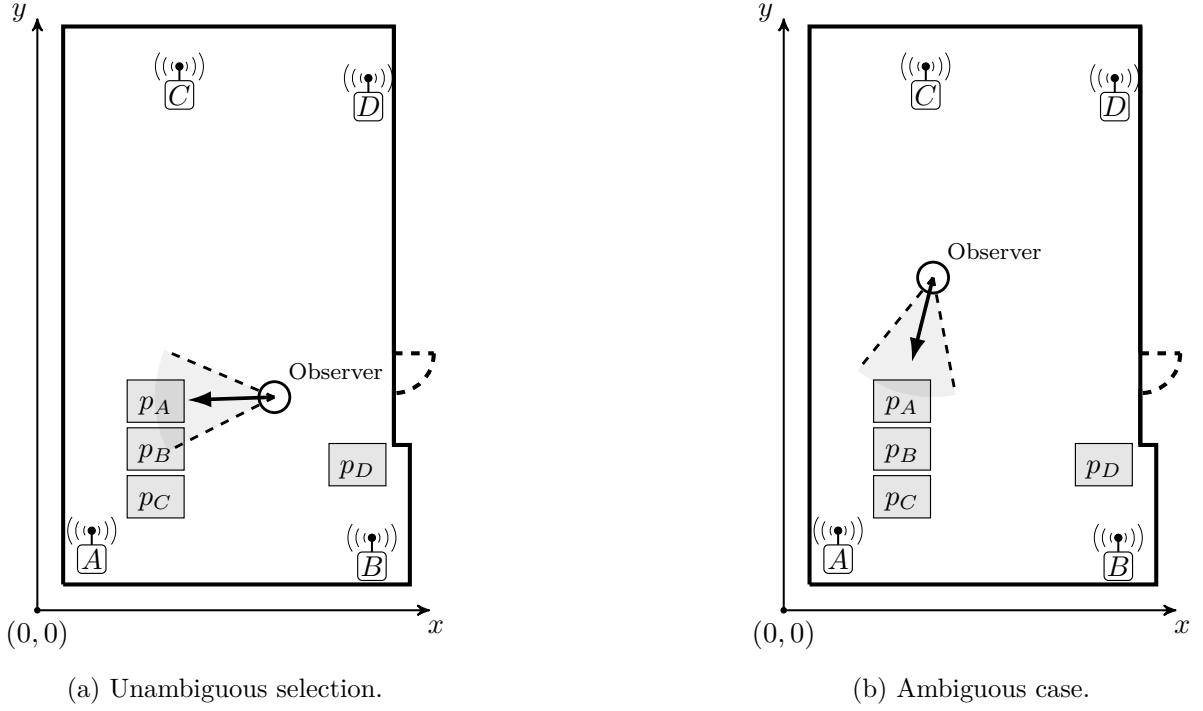


Figure 3: 2D context inference for device selection.

translation and rotation. In the near range, the reference accuracy achieved by the T-Probe is ± 0.05 mm for the 3D position, while changes in rotational angles are registered within the measurement volume with an accuracy of $\pm 0.005^\circ$. The UWB system operated at 3 Hz. All device data were temporally aligned with the laser-tracker reference in post-processing using cross-correlation of the yaw-angle time series.



Figure 4: Custom 3D-printed holder for the experiments.

6.3. Reference System and UWB Position Accuracy

The laser tracker was registered to the UWB coordinate system using four access points measured with CornerCube reflectors. The residual standard deviation of this transformation was below 10 mm, so the reference contributes negligible error to the comparison.

UWB static accuracy was assessed at eight points: mean 2D position error 0.17 m, consistent with our previous work and sufficient for coarse spatial context. During the dynamic trajectory experiment (Section 6.5), position error increased to a standard deviation of 0.47 m in X and 0.66 m in Y. This is expected: the tag moves and line-of-sight to anchors is temporarily blocked during cart movement. It does not indicate a change in the UWB system’s ranging quality. In both cases, UWB supports coarse localization but not disambiguation of closely spaced devices [3].

6.4. Sensitivity to Rotational Motion

This experiment examined how the evaluated devices respond to rotational motion. For the proposed application, slow rotations are particularly relevant, since operators often turn their heads gradually when inspecting nearby assets. Two experiments with different angular velocities were carried out.

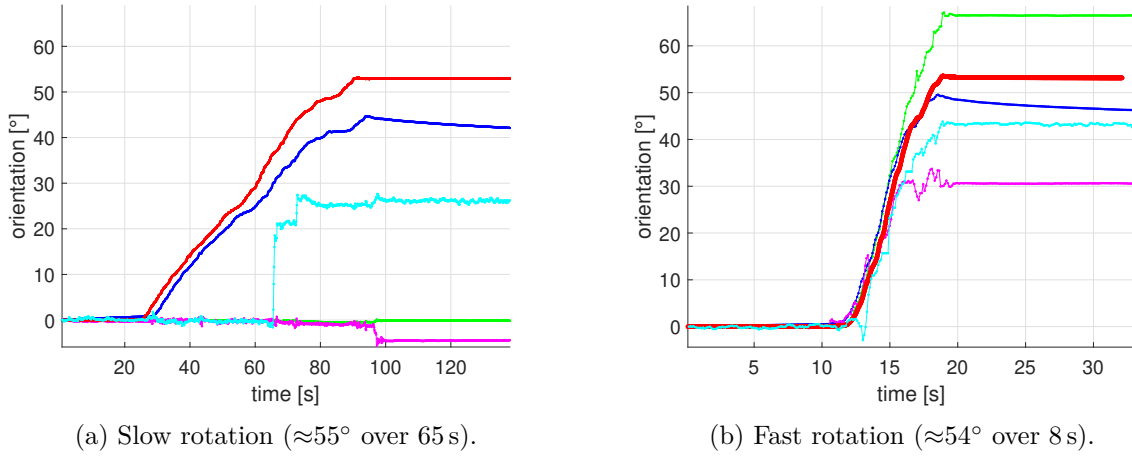


Figure 5: Yaw angle (degrees) over time (seconds) during slow and fast rotations. All devices begin from a common heading at $t=0$. Color coding: green = Google Glass, blue = Pixel 8, magenta = Vuzix Blade 2, cyan = Vuzix M4000, red = laser-tracker reference.

Figure 5 summarizes the results. In the slow experiment (Fig. 5a), Google Glass and the Vuzix Blade 2 hardly detected the motion at all, with signals remaining nearly constant despite the continuous reference rotation. The Pixel 8 captured part of the rotation but underestimated the total angle; the Vuzix M4000 showed an abrupt change rather than a continuous response.

In the fast experiment (Fig. 5b), all devices responded more clearly. Google Glass followed the rotation much better but overestimated the final angle by about 13° . The Pixel 8 tracked reasonably well with a noticeable deviation; the Vuzix Blade 2 and M4000 also responded more strongly but remained substantially below the reference trajectory.

Overall, slow rotations posed a clear problem for all evaluated devices. For practical context inference, device suitability depends strongly on how well slow head movements are captured.

6.5. Dynamic 2D Trajectory Experiment

The test body was moved through the measurement area on a cart and repeatedly rotated around the vertical axis. Figure 6 shows the UWB position estimates (blue) and the laser-tracker reference trajectory (red). The standard deviation of position differences was 0.47 m in X and 0.66 m in Y, attributable to dynamic tag motion and temporary shadowing. The results confirm that UWB supports coarse localization but not reliable disambiguation of closely spaced devices.

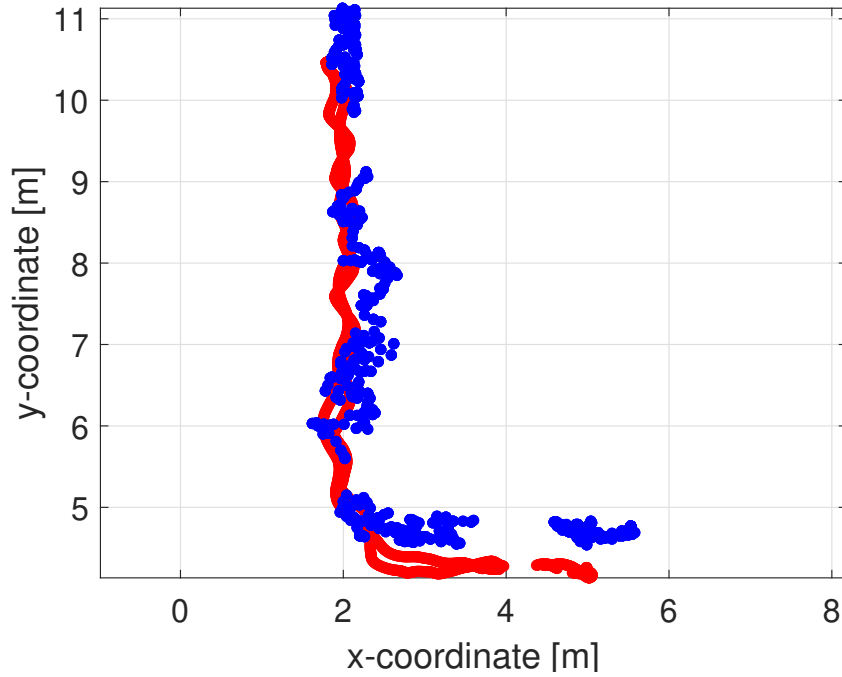


Figure 6: UWB position estimates (blue) and laser-tracker reference trajectory (red) during dynamic motion.

6.6. Exploratory Practical Test

An exploratory practical test was conducted to assess end-to-end identification in the laboratory. The test covered all four printers (p_A to p_D): from each of several test positions distributed across the workspace, the operator looked at the corresponding printer and the system was asked to identify it. All three evaluated smart glasses (Google Glass Enterprise Edition 2, Vuzix Blade 2, and Vuzix M4000) correctly identified all four printers in all tested position-device combinations. Figure 7 shows a representative case (printer p_D).

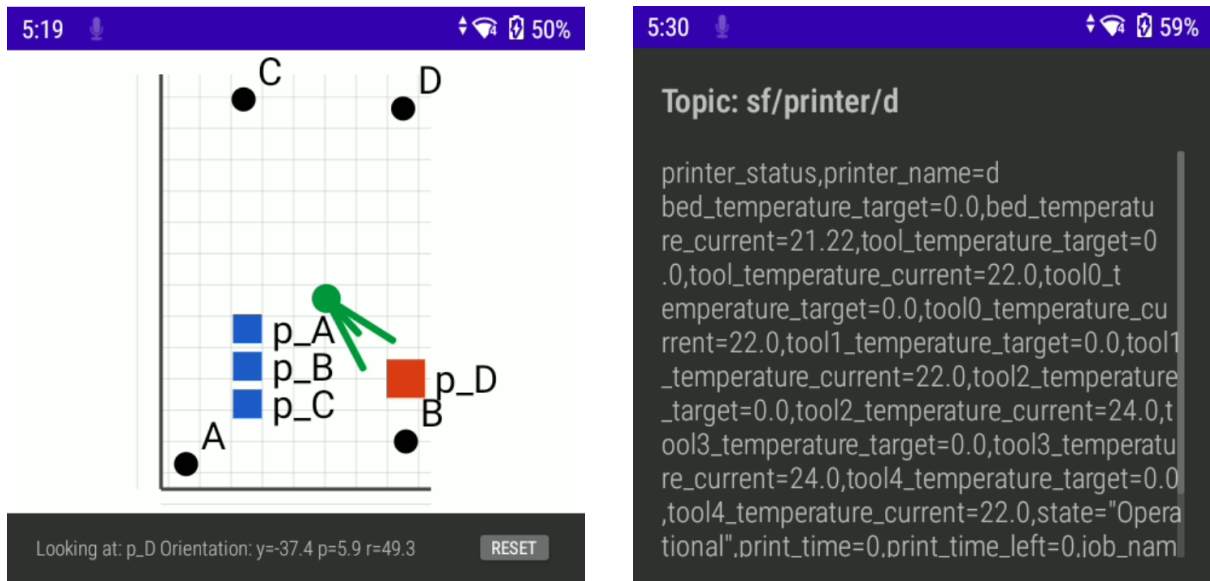


Figure 7: Smart-glass application during the practical test (printer p_D).

The prototype thus works under laboratory conditions when head rotations are fast enough. The test is exploratory: viewing distances, approach angles, and rotation speeds were not systematically varied, and all three remain subjects of the planned follow-up study.

7. Conclusion and Outlook

We presented a preliminary prototype combining low-cost UWB positioning with on-device head-orientation sensing on commodity smart glasses for workstation-level context inference. UWB provides coarse spatial context (static mean error 0.17 m, dynamic standard deviation up to 0.66 m) but is not sufficient for asset disambiguation on its own. Orientation-based disambiguation works at sufficient rotation speeds but fails for slow head movements on all tested devices, which is the primary limitation to address. In an exploratory practical test, all three smart-glasses models correctly identified all four evaluated printers. This paper reports work in progress. The next step is a structured study varying rotation speed, viewing distance, and approach angle, and evaluating fallback strategies for slow rotations.

Open-Source Artifact

We release the complete Android reference implementation and the measurement data required to reproduce the core results: github.com/thl-smart-factory-lab/look-where-you-work.

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Generative AI Disclosure

No generative AI tools were used in the writing or preparation of this paper.

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